



PIYUSH MAJI
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EDUCATION

Indian Institute of Technology Hyderabad

Expected July 2027

M.Sc. in Physics CGPA: 7.5/10

Hooghly Mohsin College (1836), West Bengal

August 2024

B.Sc. in Physics CGPA: 7.28/10

RESEARCH

GNN-Based Full Event Interpretation (GraFEI) — Belle II

Ongoing

Prof. Sourav Sandilya, Department of Physics, IIT Hyderabad

- Studying ML-driven inclusive event reconstruction for the Belle II experiment at KEK using the Graph-based Full Event Interpretation (GraFEI) framework, which applies deep GNNs to reconstruct full decay topologies from final state particles alone, without prior assumptions on the decay chain structure.
- Building detector-level graph representations of particle decay chains using Python and PyTorch Geometric; modelling each reconstructed particle as a graph node with kinematic features (momentum, DOCA, $\cos\theta$) and learning the Lowest Common Ancestor Stage (LCAS) matrix encoding the decay hierarchy ($\Upsilon(4S) \rightarrow B \rightarrow D^* \rightarrow \dots$).
- Training the GraFEIModel (stacked MetaLayers with EdgeLayer, NodeLayer, GlobalLayer) to jointly predict particle mass hypotheses and the LCAS matrix via a combined cross-entropy loss $\mathcal{L} = \mathcal{L}_{LCA} + \alpha \mathcal{L}_{Masses}$; evaluating reconstruction quality using PerfectLCA, PerfectMasses, and PerfectEvent metrics within the basf2 analysis framework.

PROJECTS

GNN-FraudNet — Transaction Fraud Detection

Completed

Python, PyTorch Geometric, GNN [GitHub](#)

- Built a Graph Neural Network pipeline on the Elliptic Bitcoin Dataset (203K nodes, 234K edges) to classify illicit transactions; modelled the transaction graph with 166-dimensional node features and temporal edge structure, framing fraud detection as a node classification task under severe class imbalance (2% illicit).
- Implemented and benchmarked GCN, GraphSAGE, and GAT architectures in PyTorch Geometric; applied class-weighted cross-entropy and SMOTE-based oversampling to handle imbalance; evaluated with precision, recall, F1, and AUC-ROC, achieving competitive illicit-class F1 on the held-out test set.

Spam Classifier from Scratch

Completed

Python, NumPy, Machine Learning [GitHub](#)

- Built Multinomial Naive Bayes (MLE + Laplace smoothing) and Logistic Regression (gradient descent) entirely from scratch in NumPy on the UCI SMS Spam Collection; implemented both TF-IDF and Bag-of-Words feature pipelines without sklearn, with vocabulary built strictly on the train split to prevent data leakage.
- Implemented binary cross-entropy loss and hand-coded gradient descent, visualising real-time loss convergence; evaluated both models with precision, recall, F1, and ROC/AUC computed via the trapezoidal rule; analysed threshold sensitivity (0.3–0.7) to explore the precision-recall tradeoff on an 86.6% class-imbalanced dataset.

COVID-19 Epidemic Modeling — SIR / SEIR / SUTRA

Completed

Prof. Anupam Gupta, IIT Hyderabad [GitHub](#)

- Implemented SIR, SEIR, and SUTRA compartmental models on US COVID-19 data (Feb 2020–Mar 2021); used NNLS regression for per-phase parameter estimation, achieving MAPE $\sim 4\%$ on individual phases; ensembled all three models via linear regression for improved generalisation.

RELEVANT COURSES

Machine Learning for Physical Sciences • Machine Learning for High Energy Physics • Statistical Mechanics • Linear Algebra • Computational Physics • Data Science Analysis

SKILLS

Languages & Tools: Python, Git, NumPy, Pandas, PyTorch, PyTorch Geometric, basf2

Domain: Machine Learning, Graph Neural Networks, Scientific Computing, Data Science, Data Structures & Algorithms

Coding Practice: LeetCode, GeeksforGeeks, Striver A2Z, Codeforces

LEADERSHIP & EXTRACURRICULAR

Placement Coordinator

May 2026 – Present

Office of Career Services, IIT Hyderabad

Managing end-to-end campus recruitment drives, liaising with recruiting companies, and overseeing placement operations for the student body. Onboarded companies like IBM, Licious, HCL, etc.

Inbound & Logistics Management

Sep 2025 – Present

International Relations Cell, IIT Hyderabad

Coordinating orientations, cultural programs, and academic tours for international exchange students from Japan and the United States; managing logistics for visiting international guests.

Design Core

May 2026 – Present

Extra Mural Lectures (EML), IIT Hyderabad

Leading visual design and multimedia outreach for EML; creating promotional content, event branding, and design assets for club initiatives and student entrepreneurship programs. Hosted NASA Axiom Mission Pilot Subhanshu Shukla.